

Forecasting Teacher Supply in Tanzania

Your Name

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Overview & Motivation

- Supply-side teacher forecasting framework for Tanzania
- Supports MOEST and PO-RALG in workforce planning
- Motivated by:
 - Rapid expansion of education access (fee-free, curriculum reforms)
 - Persistent supply challenges: STEM shortages, uneven distribution, attrition
- Central idea: forecast systematically, not reactively

Core Concept: Teacher Supply

- Teacher supply = Current stock + Inflows – Outflows
- Influenced by:
 - Individual decisions (career choice, mobility)
 - Government policies (training, recruitment, compensation)

Purpose of the Framework

- Transparent and repeatable forecasting system
- Supports planning, budgeting, recruitment decisions
- Produces disaggregated forecasts (subject, gender, region, level)
- Aligns teacher supply with education demand

Guiding Principles

- Policy relevance (aligned with ESDP, national strategies)
- Institutionalization (routine, not ad hoc)
- Data-driven decision-making
- Simplicity with scalability
- Equity and inclusion (gender, rural, special needs)

Institutional Framework

- Coordination among:
 - MOEST → policy & technical leadership
 - PO-RALG → deployment & local implementation
 - Training institutions & TCU → supply pipeline
 - NBS & Ministry of Finance → demographics & budgeting
- Technical working group proposed for coordination

Forecasting Methodology

- **Stock-and-Flow Model**

$$[\text{Stock}_{\{t+1\}} = \text{Stock}_t + \text{Inflows}_t - \text{Outflows}_t]$$

- Advantages:

- Requires minimal data
- Transparent and easy to implement
- Can be updated regularly

Model Components & Supporting Methods

Inflows

- New graduates entering teaching
- Re-entry of former teachers

Outflows

- Retirement (predictable)
- Mortality
- Attrition (resignation, migration, career change)

Supporting Methods

Data Constraints & Implementation Strategy

- Tanzania lacks longitudinal, high-quality microdata
- Administrative systems (EMIS, HR, payroll) useful but incomplete
- Three-phase roadmap:
 - 1 Pilot – data integration & initial model
 - 2 Scale-up – national implementation
 - 3 Enhancement – advanced methods

Cross-Cutting Priorities

- Gender balance
- Regional equity (urban vs rural)
- Special needs education
- Data governance & quality assurance

Final Insight

- Framework shifts planning from:
 - Reactive → responding to shortages
 - Proactive → anticipating future supply
- Teacher forecasting is a strategic policy tool
- Ensures every learner has access to a qualified, well-distributed, motivated teacher